AP650

LCD AND KEYBOARD CONTROLLER

Datasheet

Table of Contents

1 GENERAL DESCRIPTION 1

2 FEATURES 1

3 ORDERING INFORMATION 1

4 PIN CONFIGURATION (LQFP64) 2

5 BLOCK DIAGRAM 2

6 PIN DESCRIPTION 3

7 FUNCTIONAL DESCRIPTION 4

7.1 LCD DRIVERS 5
7.2 PORT 1 AND PORT 2 7
7.3 PORT 3 8
7.4 ANALOG-TO-DIGITAL CONVERSION 9
7.5 SET REG 11
7.6 KEYBOARD SCANNER 12

8 SERIAL INTERFACE FORMAT 14

9 SERIAL COMMAND SETS 16

10 LCD TIMING WAVEFORMS 17

11 ELECTRICAL SPECIFICATION 18

11.1 ABSOLUTE MAXIMUM RATING 18
11.2 RECOMMENDED OPERATING CONDITION 18
11.3 ELECTRICAL CHARACTERISTICS 18

12 PACKAGE INFORMATION 19

12.1 BONDING INFORMATION 19
12.2 COB PACKAGE 20
12.3 LQFP-64 20

Figures

Figure 1 64-Pin LQFP Package of AP650 2
Figure 2: Block Diagram of LCD AND KEYBOARD CONTROLLER 2
Figure 3: LCD Display Data Structure Setting 5
Figure 4: Example of Display Mask Enable 5
Figure 5: LCD Display RAM 5
Figure 6: Structure of the Port 1 and Port 2 pins 7
Figure 7: Structure of the Port 3 8
Figure 8: Analog-to-digital Converter Block 9
Figure 9: Automatic Mode Timing 10
Figure 10: Keyboard Matrix 12
Figure 11: Serial Communication Format 14
Figure 12: Host transfer data to LCD AND KEYBOARD CONTROLLER 14
Figure 13: Host Request Data from LCD AND KEYBOARD CONTROLLER 14
Figure 14: Serial Communication Timing 15
Figure 15: Serial Communication Data Format 16
Figure 16: LCD waveforms 17

1 General Description

LCD AND KEYBOARD CONTROLLER is a controller IC with built-in LCD drivers and keyboard scanners. It supports a maximum of 40 segments and 4 commons with a maximum of 24 I/O pins.

2 Features

- Built-in 1.5V voltage regulator for stabilizing the LCD supply voltage
- Built-in 40x4 1/3 bias, 1/4 duty LCD drivers
- Supports a maximum of 160 LCD display dots with 40 segments and 4 commons
- Support external LCD supply voltage
- 16 output port pins and 8 input port pins
- 8 channels 6-bit Analog-to-digital converter
- 2 pins serial interface from external host controller
- Built-in RC oscillator
- Low operating voltage: 1.8V to 5.5V

3 Ordering Information

Ordering Number	Pins	Package
AP650-LQ-L	64	LQFP
AP650-DC-L	64	Dice Form
AP650-CB-L	64	COB

4 Pin Configuration (LQFP64)

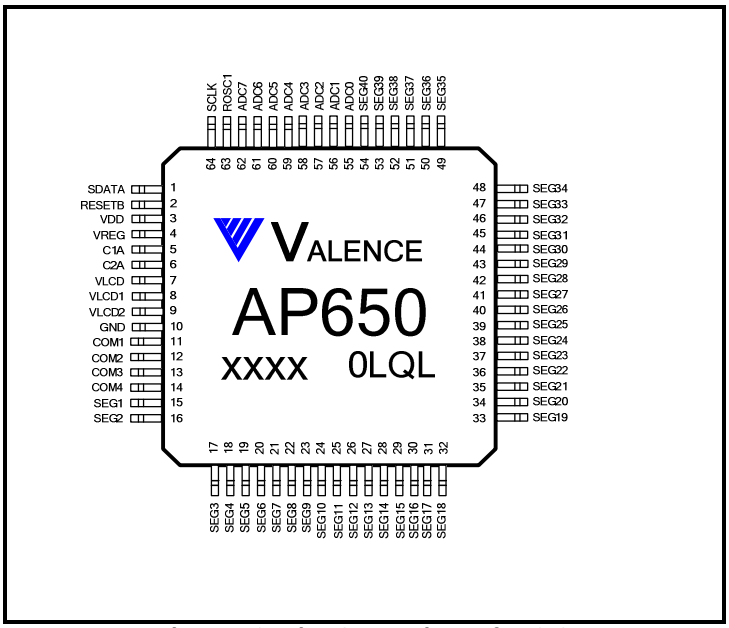


Figure 1 64-Pin LQFP Package of AP650

5 Block Diagram

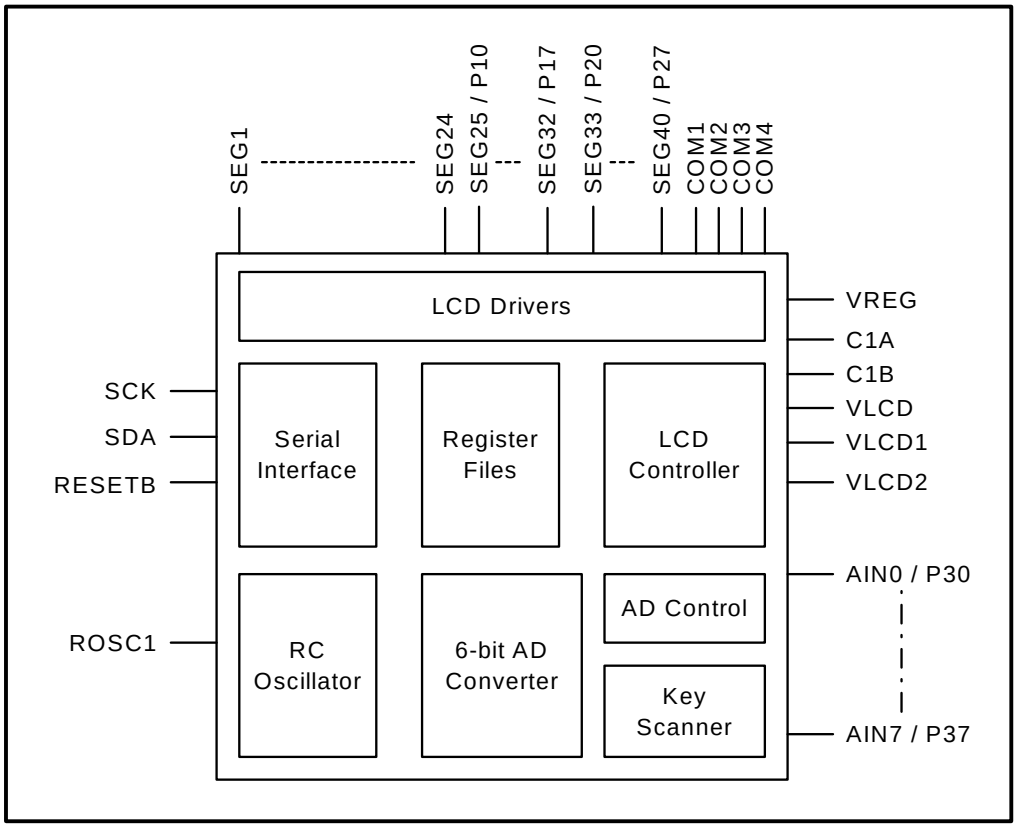


Figure 2: Block Diagram of LCD AND KEYBOARD CONTROLLER

6 Pin Description

COB/Dice #	LQFP #	Name	Type	Description
1.	32.	SEG18	LCDP	LCD Segment Driver
2.	33.	SEG19	LCDP	LCD Segment Driver
3.	34.	SEG20	LCDP	LCD Segment Driver
4.	35.	SEG21	LCDP	LCD Segment Driver
5.	36.	SEG22	LCDP	LCD Segment Driver
6.	37.	SEG23	LCDP	LCD Segment Driver
7.	38.	SEG24	LCDP	LCD Segment Driver
8.	39.	SEG25 / P10	LCDP/IO	LCD Segment Driver or General Purpose Output Port
9.	40.	SEG26 / P11	LCDP/IO	LCD Segment Driver or General Purpose Output Port
10.	41.	SEG27 / P12	LCDP/IO	LCD Segment Driver or General Purpose Output Port
11.	42.	SEG28 / P13	LCDP/IO	LCD Segment Driver or General Purpose Output Port
12.	43.	SEG29 / P14	LCDP/IO	LCD Segment Driver or General Purpose Output Port
13.	44.	SEG30 / P15	LCDP/IO	LCD Segment Driver or General Purpose Output Port
14.	45.	SEG31 / P16	LCDP/IO	LCD Segment Driver or General Purpose Output Port
15.	46.	SEG32 / P17	LCDP/IO	LCD Segment Driver or General Purpose Output Port
16.	47.	SEG33 / P20	LCDP/IO	LCD Segment Driver or General Purpose Output Port
17.	48.	SEG34 / P21	LCDP/IO	LCD Segment Driver or General Purpose Output Port
18.	49.	SEG35 / P22	LCDP/IO	LCD Segment Driver or General Purpose Output Port
19.	50.	SEG36 / P23	LCDP/IO	LCD Segment Driver or General Purpose Output Port
20.	51.	SEG37 / P24	LCDP/IO	LCD Segment Driver or General Purpose Output Port
21.	52.	SEG38 / P25	LCDP/IO	LCD Segment Driver or General Purpose Output Port
22.	53.	SEG39 / P26	LCDP/IO	LCD Segment Driver or General Purpose Output Port
23.	54.	SEG40 / P27	LCDP/IO	LCD Segment Driver or General Purpose Output Port
24.	55.	ADC0 / P30	IO/IA	6-bit ADC input terminal 0 or General Purpose Input Port
25.	56.	ADC1 / P31	IO/IA	6-bit ADC input terminal 1 or General Purpose Input Port
26.	57.	ADC2 / P32	IO/IA	6-bit ADC input terminal 2 or General Purpose Input Port
27.	58.	ADC3 / P33	IO/IA	6-bit ADC input terminal 3 or General Purpose Input Port
28.	59.	ADC4 / P34	IO/IA	6-bit ADC input terminal 4 or General Purpose Input Port
29.	60.	ADC5 / P35	IO/IA	6-bit ADC input terminal 5 or General Purpose Input Port
30.	61.	ADC6 / P36	IO/IA	6-bit ADC input terminal 5 or General Purpose Input Port
31.	62.	ADC7 / P37	IO/IA	6-bit ADC input terminal 5 or General Purpose Input Port
32.	63.	ROSC1	-	RC oscillator connection terminal
33.	64.	SCLK	I	Serial Communication Clock
34.	1.	SDATA	IO	Serial Communication Data
35.	2.	RESETB	ISU	Active Low chip reset signal
36.	3.	VDD	LCDP	Voltage Supply
37.	4.	VREG	LCDP	1.5V Voltage Regulator Output. Connect a 0.1uF Capacitor from this pin to GND
38.	5.	C1A	LCDP	Voltage Double Capacitor connection terminal 1
39.	6.	C2A	LCDP	Voltage Double Capacitor connection terminal 2
40.	7.	VLCD	LCDP	LCD driver supply voltage. Connect a 0.1uF capacitor from this pin to GND.
41.	8.	VLCD1	LCDP	2/3 x VLCD voltage. Connect a 0.1uF capacitor from this pin to GND.
42.	9.	VLCD2	LCDP	1/3 x VLCD voltage. Connect a 0.1uF capacitor from this pin to GND.
43.	10.	GND	LCDP	Ground
44.	11.	COM1	LCDP	LCD Common Driver
45.	12.	COM2	LCDP	LCD Common Driver
46.	13.	COM3	LCDP	LCD Common Driver
47.	14.	COM4	LCDP	LCD Common Driver
48.	15.	SEG1	LCDP	LCD Segment Driver
49.	16.	SEG2	LCDP	LCD Segment Driver
50.	17.	SEG3	LCDP	LCD Segment Driver
51.	18.	SEG4	LCDP	LCD Segment Driver
52.	19.	SEG5	LCDP	LCD Segment Driver
53.	20.	SEG6	LCDP	LCD Segment Driver
54.	21.	SEG7	LCDP	LCD Segment Driver
55.	22.	SEG8	LCDP	LCD Segment Driver
56.	23.	SEG9	LCDP	LCD Segment Driver
57.	24.	SEG10	LCDP	LCD Segment Driver
58.	25.	SEG11	LCDP	LCD Segment Driver
59.	26.	SEG12	LCDP	LCD Segment Driver
60.	27.	SEG13	LCDP	LCD Segment Driver
61.	28.	SEG14	LCDP	LCD Segment Driver
62.	29.	SEG15	LCDP	LCD Segment Driver
63.	30.	SEG16	LCDP	LCD Segment Driver
64.	31.	SEG17	LCDP	LCD Segment Driver

I: Input pad ISU: CMOS Schmitt trigger input with pull-up IA: Analog input
LCDP: LCD pad IO: Input / Output pad

7 Functional Description

A set of register files together with 40bytes display RAM are built inside LCD AND KEYBOARD CONTROLLER for controlling the operation modes and functionality of the chip. Below is a summary of the register files in AP650. The functionality of each register file is described in the following sub-sections.

Address	Name	R/W Status	Description
00H	SEGCFG1	WO	Segment Configuration Register 1
01H	SEGCFG2	WO	Segment Configuration Register 2
02H	SEGCFG3	WO	Segment Configuration Register 3
03H	SEGCFG4	WO	Segment Configuration Register 4
04H	LCDCTR	WO	LCD Control Register
05H	P1D	WO	Port 1 Data Register
06H	P2D	WO	Port 2 Data Register
07H	P3D	RO	Port 3 Data Register
08H	P3PUEN	WO	Port 3 Pull-up Resistor Enable register
09H	ADCFG	WO	Analog Input Pin configuration Register
0AH	SETREG	WO	Set Register
0BH	ADCTR	R/W	AD Mode Control Register
0CH	KSCTR	R/W	Key Scan Control Register
10H	ADCH0	RO	AD Channel 0 Data or Keyboard Scan Data for P20 output
11H	ADCH1	RO	AD Channel 1 Data or Keyboard Scan Data for P21 output
12H	ADCH2	RO	AD Channel 2 Data or Keyboard Scan Data for P22 output
13H	ADCH3	RO	AD Channel 3 Data or Keyboard Scan Data for P23 output
14H	ADCH4	RO	AD Channel 4 Data or Keyboard Scan Data for P24 output
15H	ADCH5	RO	AD Channel 5 Data or Keyboard Scan Data for P25 output
16H	ADCH6	RO	AD Channel 6 Data or Keyboard Scan Data for P26 output
17H	ADCH7	RO	AD Channel 7 Data or Keyboard Scan Data for P27 output
80H – FFH		R/W	LCD RAM Addresses

7.1 LCD Drivers

LCD AND KEYBOARD CONTROLLER supports 40x4 LCD display dots with built-in 80x4 display RAM. The display RAM is organized in 4-bit format with address ranges from 80H to 0FFH which corresponds to 160 display dots together with another 160 bits Display Mask Enable Data. Each display dot works together with the Display Mask Enable bit. The Display Mask Enable Bit is used to control the display dot together with the MASK_ON bit on bit 6 of the LCD control register. When both the MASK_ON bit and the Display Mask Enable are set to logic one for a LCD display dot, the corresponding LCD display dot will be turned OFF, independent of the setting of the LCD display RAM setting. Figure 5 below shows the relationship between the LCD RAM and the LCD segment and common signals and below shows an example of the Display Mask Enable bit.

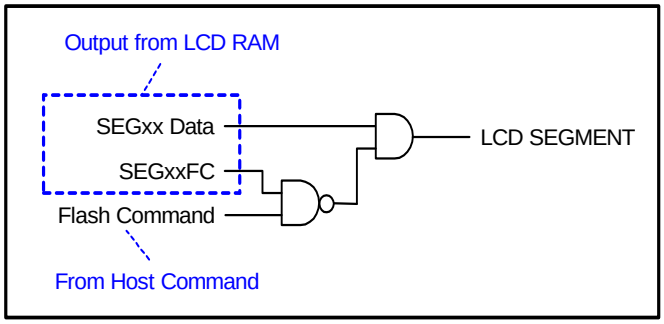


Figure 3: LCD Display Data Structure Setting

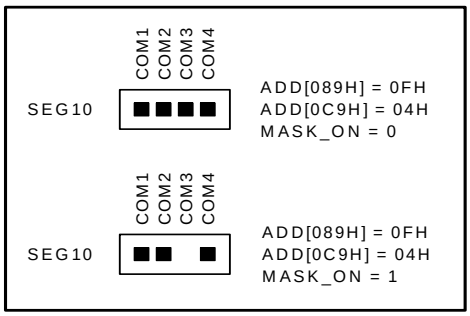


Figure 4: Example of Display Mask Enable

	COM4	COM3	COM2	COM1		COM4	COM3	COM2	COM1
Address	Bit 3	Bit 2	Bit 1	Bit 0	Address	Bit 3	Bit 2	Bit 1	Bit 0
80H	SEG1				C0H	SEG1FC			
81H	SEG2				C1H	SEG2FC			
82H	SEG3				C2H	SEG3FC			
83H	SEG4				C3H	SEG4FC			
84H	SEG5				C4H	SEG5FC			
85H	SEG6				C5H	SEG6FC			
86H	SEG7				C6H	SEG7FC			
87H	SEG8				C7H	SEG8FC			
88H	SEG9				C8H	SEG9FC			
89H	SEG10				C9H	SEG10FC			
8AH	SEG11				CAH	SEG11FC			
8BH	SEG12				CBH	SEG12FC			
8CH	SEG13				CCH	SEG13FC			
8DH	SEG14				CDH	SEG14FC			
8EH	SEG15				CEH	SEG15FC			
8FH	SEG16				CFH	SEG16FC			
90H	SEG17				D0H	SEG17FC			
91H	SEG18				D1H	SEG18FC			
92H	SEG19				D2H	SEG19FC			
93H	SEG20				D3H	SEG20FC			
94H	SEG21				D4H	SEG21FC			
95H	SEG22				D5H	SEG22FC			
96H	SEG23				D6H	SEG23FC			
97H	SEG24				D7H	SEG24FC			
98H	SEG25				D8H	SEG25FC			
99H	SEG26				D9H	SEG26FC			
9AH	SEG27				DAH	SEG27FC			
9BH	SEG28				DBH	SEG28FC			
9CH	SEG29				DCH	SEG29FC			
9DH	SEG30				DDH	SEG30FC			
9EH	SEG31				DEH	SEG31FC			
9FH	SEG32				DFH	SEG32FC			
A0H	SEG33				E0H	SEG33FC			
A1H	SEG34				E1H	SEG34FC			
A2H	SEG35				E2H	SEG35FC			
A3H	SEG36				E3H	SEG36FC			
A4H	SEG37				E4H	SEG37FC			
A5H	SEG38				E5H	SEG38FC			
A6H	SEG39				E6H	SEG39FC			
A7H	SEG40				E7H	SEG40FC			

Figure 5: LCD Display RAM

The lower 8 LCD segment pin, SEG1 to SEG8 and the upper 16 LCD segment pins, SEG33 to SEG40, can be configured flexibly as LCD driver terminals, keyboard scanner output terminals or general purpose output terminals. Each individual pin can be configured independently by the Segment Configuration Register 1, the Segment Configuration Register 2, the Segment Configuration Register 3 and the Segment Configuration Register 4.

Segment Configuration Register 1 (Register Address : 00H, WO)

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
SEG28CFG[1:0]		SEG27CFG[1:0]		SEG26CFG[1:0]		SEG25CFG[1:0]	
0	0	0	0	0	0	0	0

Segment Configuration Register 2 (Register Address : 01H, WO)

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
SEG32CFG[1:0]		SEG31CFG[1:0]		SEG30CFG[1:0]		SEG29CFG[1:0]	
0	0	0	0	0	0	0	0

Segment Configuration Register 3 (Register Address : 02H, WO)

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
SEG36CFG[1:0]		SEG35CFG[1:0]		SEG34CFG[1:0]		SEG33CFG[1:0]	
0	0	0	0	0	0	0	0

Segment Configuration Register 4 (Register Address : 03H, WO)

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
SEG40CFG[1:0]		SEG39CFG[1:0]		SEG38CFG[1:0]		SEG37CFG[1:0]	
0	0	0	0	0	0	0	0

These four registers are used to configure the functionality of the SEG25 to SEG40 pins

SEGxxCFG[1:0] Will configure function of the SEGxx Pin
00B – Disable the pin. The pin will be in tri-state
01B – Pin set as LCD segment output
10B – Pin set as keyboard scan output (for SEG33 to SEG40 only and is reserved for SEG25 to SEG32)
11B – Pin set as general purpose output port

LCD Control Register (Register Address : 04H, WO)

This register is used to control the functionality of the LCD Controller

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	MASK_EN	HI_DRV	BIAS0	BIAS1	REGEN	LCDCLR	LCDEN
0	0	0	0	0	0	1	0

LCDEN This bit is used to enable / disable the LCD display
0 – disable LCD display controller function
1 – enable LCD display controller function

LCDCLR Setting of this bit will cause the LCD display to blank without clearing the content of the display RAM. The LCD controller still in the scanning mode
0 – enable display data to come out to driver
1 – blank all display data

REGEN This bit is used to enable / disable the LCD regulator. Disabling the LCD regulator will also disable the voltage doubler
0 – disable LCD regulator and doubler (VDD can be connected externally to VLCD pin as LCD supply voltage)
1 – enable LCD regulator and doubler

BIAS1 Setting this bit to “1” will improve the driving capability of the VLCD1 and VLCD2
BIAS0 Setting this bit to “1” will enable the VLCD1 and VLCD2, otherwise VLCD1 and VLCD2 will drive to “0”

BIAS1	BIAS0	Description
0	0	VLCD1,2 internal biasing are OFF
0	1	VLCD1,2 internal biasing are ON and in low driving mode
1	0	Not recommended
1	1	VLCD1,2 internal biasing are ON and in high driving mode

MASK_EN This bit is used to set the enable / disable the function of the LCD segment Mask bit
0 – disable the function of the SEGFC bit. That is, the display dot will accord to the setting of the SEG data
1 – enable the function of the SEGFC bit. The display dot will be mask off if the SEGFC bit is “1”
HI_DRV Setting this bit to “1” will improve the driving capability of the LCD COM/SEG pin drivers.
Bit7 Reserved. Should always write ‘0’ to this bit.

7.2 Port 1 and Port 2

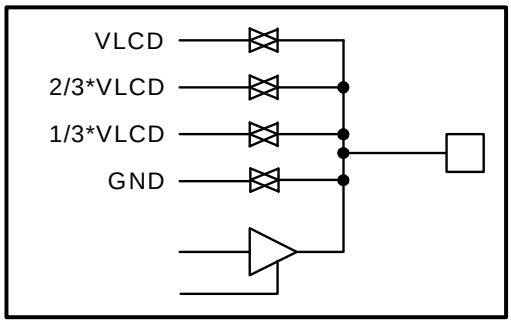


Figure 6: Structure of the Port 1 and Port 2 pins

Figure 6 above shows the structure of the SEG1 to SEG8 and SEG33 to SEG40 pins. SEG1 to SEG8 are shared with the P10 to P17 pins and SEG33 to SEG40 shares with P20 to P27 pins. The selection of the usage of the SEG1 to SEG8 and SEG33 to SEG40 pins are through the Segment Configuration Registers 1 to 4. When the pin is configured as general output port pin, the following register will be used to output data to the port 1 and port 2 pins.

Port 1 Data Register (Register Address : 05H, WO)

This register is used to output data to P10 to P17 pins if they are configured as port data output. The data written to this register will not have effect to those port pins which are not configured as data output port.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
P17	P16	P15	P14	P13	P12	P11	P10
0	0	0	0	0	0	0	0

Port 2 Data Register (Register Address : 06H, WO)

This register is used to output data to P20 to P27 pins if they are configured as port data output. The data written to this register will not have effect to those port pins which are not configured as data output port.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
P27	P26	P25	P24	P23	P22	P21	P20
0	0	0	0	0	0	0	0

7.3 Port 3

Port 3 is an 8-bit input port which shares with the analog channel input pins AIN0 to AIN7. The selection of the port 3 pins for analog channel input or for digital data input is controlled by the AD Configuration Register (ADCFG). Figure 7 below shows the structure of each port 3 pin. When the port is configured as analog input pin, signal will route through the analog switch to the 6-bit ADC. When the port is configured as digital input pin, the analog switch will be disabled permanently and the AND gate will be enabled. The pull up resistor can also be enabled by the Port 3 Pull-up Control Register.

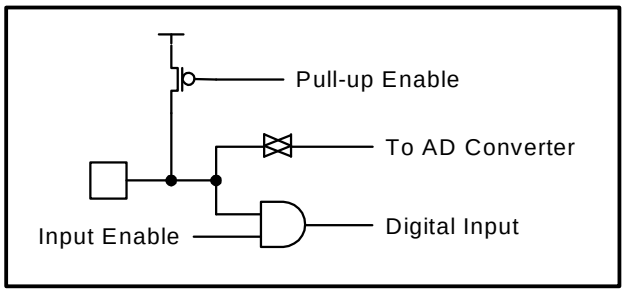


Figure 7: Structure of the Port 3

Port 3 Data Register (Register Address : 07H, RO)

This register is used to read the data input from Port 3.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
P37	P36	P35	P34	P33	P32	P31	P30
RO	RO	RO	RO	RO	RO	RO	RO

Port 3 Pull-up Control Register (Register Address : 08H, WO)

This register is used to enable / disable the pull-up resistor connected to the port 3. The bit in this register is valid only if the port pins are configured as digital input pin.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
P37_PUEN	P36_PUEN	P35_PUEN	P34_PUEN	P33_PUEN	P32_PUEN	P31_PUEN	P30_PUEN
0	0	0	0	0	0	0	0

P30_PUEN	0 – disable P30 pull-up resistor	1 – enable P30 pull-up resistor
P31_PUEN	0 – disable P31 pull-up resistor	1 – enable P31 pull-up resistor
P32_PUEN	0 – disable P32 pull-up resistor	1 – enable P32 pull-up resistor
P33_PUEN	0 – disable P33 pull-up resistor	1 – enable P33 pull-up resistor
P34_PUEN	0 – disable P34 pull-up resistor	1 – enable P34 pull-up resistor
P35_PUEN	0 – disable P35 pull-up resistor	1 – enable P35 pull-up resistor
P36_PUEN	0 – disable P36 pull-up resistor	1 – enable P36 pull-up resistor
P37_PUEN	0 – disable P37 pull-up resistor	1 – enable P37 pull-up resistor

7.4 Analog-to-Digital Conversion

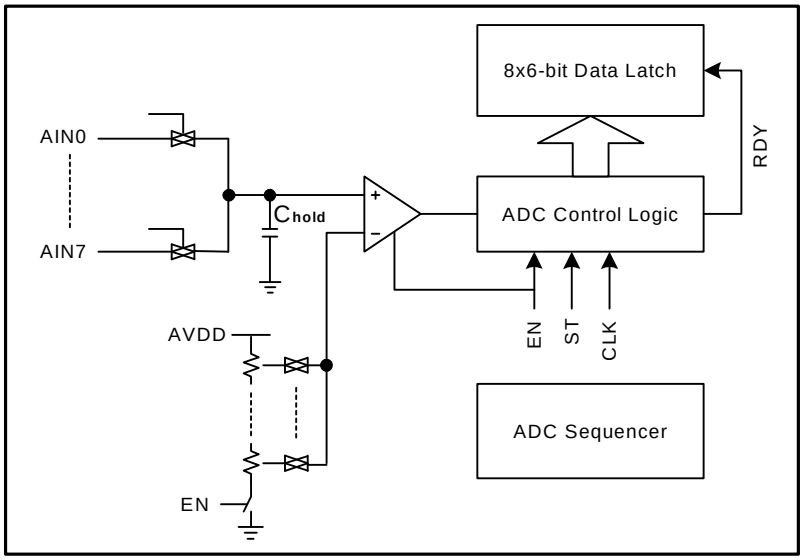


Figure 8: Analog-to-digital Converter Block

There are 8 analog input channels which can input data to the internal 6-bit analog to digital converter. These 8 analog channels share the same physical pins with the Port 3. The selection for the use of the pins are from the AD Configuration Register (ADCFG).

AD Configuration Register (Register Address : 09H, WO)

This register is used to configure the usage of the AIN/P3 pins.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
AIN7_CFG	AIN6_CFG	AIN5_CFG	AIN4_CFG	AIN3_CFG	AIN2_CFG	AIN1_CFG	AIN0_CFG
0	0	0	0	0	0	0	0

AIN0_CFG This bit will be used to configure the operation of the AIN0 / P30 pin.
0 – configure AIN0 / P30 pin as analog input channel 0
1 – configure AIN0 / P30 pin as digital data input pin and enable the setting of the P30_PUEN bit

AIN1_CFG This bit will be used to configure the operation of the AIN1 / P31 pin.
0 – configure AIN1 / P31 pin as analog input channel 1
1 – configure AIN1 / P31 pin as digital data input pin and enable the setting of the P31_PUEN bit

AIN2_CFG This bit will be used to configure the operation of the AIN2 / P32 pin.
0 – configure AIN2 / P32 pin as analog input channel 2
1 – configure AIN2 / P32 pin as digital data input pin and enable the setting of the P32_PUEN bit

AIN3_CFG This bit will be used to configure the operation of the AIN3 / P33 pin.
0 – configure AIN3 / P33 pin as analog input channel 3
1 – configure AIN3 / P33 pin as digital data input pin and enable the setting of the P33_PUEN bit

AIN4_CFG This bit will be used to configure the operation of the AIN4 / P34 pin.
0 – configure AIN4 / P34 pin as analog input channel 4
1 – configure AIN4 / P34 pin as digital data input pin and enable the setting of the P34_PUEN bit

AIN5_CFG This bit will be used to configure the operation of the AIN5 / P35 pin.
0 – configure AIN5 / P35 pin as analog input channel 5
1 – configure AIN5 / P35 pin as digital data input pin and enable the setting of the P35_PUEN bit

AIN6_CFG This bit will be used to configure the operation of the AIN6 / P36 pin.
0 – configure AIN6 / P36 pin as analog input channel 6
1 – configure AIN6 / P36 pin as digital data input pin and enable the setting of the P36_PUEN bit

AIN7_CFG This bit will be used to configure the operation of the AIN7 / P37 pin.
0 – configure AIN7 / P37 pin as analog input channel 7
1 – configure AIN7 / P37 pin as digital data input pin and enable the setting of the P37_PUEN bit

Automatic Conversion Mode

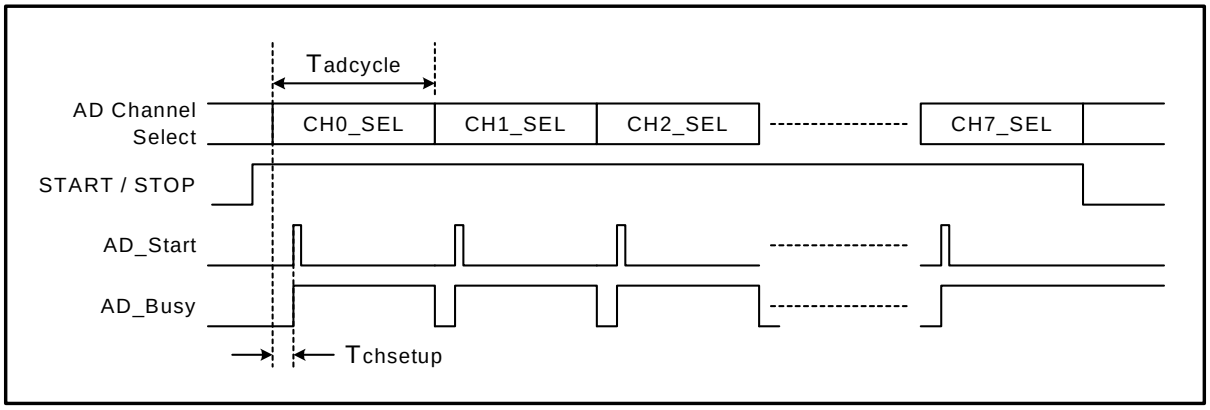


Figure 9: Automatic Mode Timing

The ADC block has a built-in sequencer which provides an automatic conversion mode for the enabled analog channels. Figure 9 below shows the timing of the automatic conversion mode. In this mode, each enabled channel (with the AIN_EN bit in the AD Channel Enable Register is "1") will be selected for a period of Tadcycle time (which is about 1ms – 2ms) [1] to perform the AD conversion and the converted data will be stored into the AD Channel Data Register. After the setting of the START / STOP bit in the AD Mode Control Register, the sequencer will start function until all the enabled AD channels have completed a conversion.
For each selected AIN channel, a delay of Tchsetup (about 200μs – 400μs) [2] will be exercised before starting the AD conversion. This is to ensure that the internal sample and hold circuit has sufficient time to charge up to the selected channel voltage. After completing the AD conversion, the data will be latched into the AD Channel Data Register and next enabled channel will be selected. Only those enabled AD channel will be selected to perform conversion in the automatic mode. After all the enabled channels have performed the AD conversion, the sequencer will stop automatically and the START / STOP bit in the AD Mode Control Register will be reset to "0".

Note [1],[2] : Tosc = 1/32768Hz is assumed

Manual Conversion Mode

Under this mode, the AD sequencer will not start conversion until receiving the conversion start command from the host. Once the conversion start command is received, AD conversion will start on the selected channel. As in the automatic mode, Tadcycle time is required to complete a conversion for a channel and Tchsetup time will be required after receiving the start conversion command and actual start of the AD conversion. The status of the AD conversion can be read back from bit 7 of the AD data register. When this bit is set, it means that AD conversion is still in progress. After completion of the conversion, the converted data will be saved into the AD data register and bit 7 of this register will also be cleared.

AD Mode Control Register (Register Address : 0BH, R/W)

This register is used to control the operation mode of the ADC and also used to select the AD channels in manual operation mode

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
AD_EN	MODE	START / STOP	Reserved	Reserved	CH2	CH1	CH0
0	0	0	0	0	0	0	0

AD_EN This bit master enable bit for the ADC block. When this bit is "0", the AD sequencer, the AD Converter will be in the reset state. All other bits in this register will not have function..
0 – disable AD conversion Block function and put it in power down state
1 – enable AD conversion Block and accept settings in the AD Mode Control Register

MODE This bit is used to set the operation mode of the AD conversion block and will be valid if AD_EN is set to "1".
0 – AD conversion Block operates in manual mode (only the channel selected by CH[2:0] would be processed).
1 – AD conversion Block operates in automatic mode (all 8 AD channels would be processed).

START / STOP This bit is used to start or stop the AD conversion
0 – Stop AD conversion after the current conversion cycle
1 – This bit is used to start the AD conversion cycle. Setting this bit will start the conversion cycles. After completion of the AD conversion, this bit will go back to '0' automatically and AD conversion will not start until next rising edge of this bit.

CH[2:0] These 3 bits are used to set the channel to be converted in manual operation mode, and the channel data be read by the Read ADC data command (serial communication command).

Bit4, 3 Reserved. Should always write '0' to these bits.

AD Channel Data Register (Register Address : 010H – 017H, RO)

These 8 registers will hold the AD converted data

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
AD7	AD6	AD5	AD4	AD3	AD2	AD1	AD0
0	0	0	0	0	0	0	0

7.5 SET REG

SET Register (Register Address: 0AH, WO)

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	Reserved	OFF_OSC	REF_OUT	REG_HDR	VDOU_HDR	TST1	TST0
		0	0	0	0	0	0

OFF_OSC: This bit will be used to turn on or off the RC oscillator
1 — turn off the oscillator
0 — turn on the oscillator

REF_OUT: Setting this bit to "1" will be set the regulator output to the AD converter . (The AD should be set to manual mode)

REG_HDR: Setting this bit to "1" will improve the driving capability of the regulator.

VDOU_HDR: Setting this bit to "1" will improve the driving capability of the doubler.

TST1: This bit will be used to configure the output port 1 and port2 to Segment data out
1 — port 1 and port 2 set to output Segment data in logic level
0 — port 1 and port 2 output data from Data register
(The SEGCFG should be set to general purpose output port first)

TST0: This bit will be used to configure the output of COM[3:0] and SEG[24:1] to logic level
1 — Set to logic output
0 — Set to VLCD output

7.6 Keyboard Scanner

LCD AND KEYBOARD CONTROLLER supports a hardware keyboard scanning sequencer, which uses P20 to P27 as the keyboard output pins and P3 as keyboard input pins. So a maximum of 8x8 key matrix can be formed. Figure 10 below shows the key matrix.

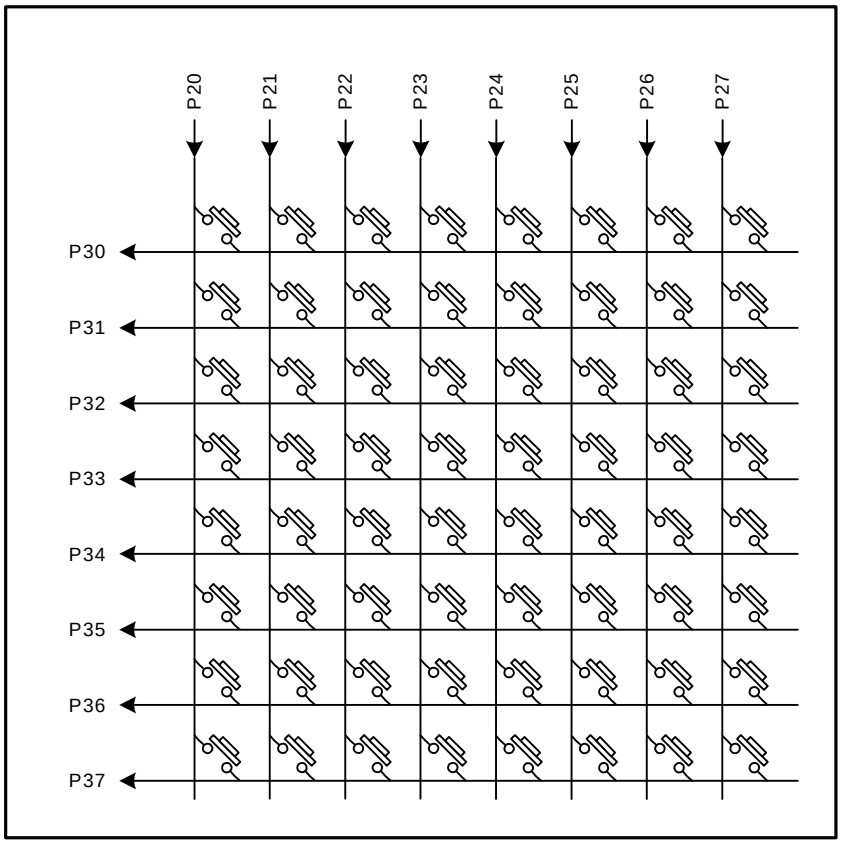


Figure 10: Keyboard Matrix

The keyboard scan sequencer operates in 2 modes, namely the manual scan mode or the automatically scan mode. In manual scan mode, host has to output a data to the P2 and then read back data from P3. In automatic scan mode, those P2 pins which have been configured for Keyboard Scanning function will be scanned once and the data will be latched from P3 into the Key Scan Data Register. After completion of the scan cycle, the sequencer will stop until next START / STOP command is received.

Key Scan Control Register (Register Address : 0CH, R/W)

This register is used to control the operation of the keyboard scanner

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
KS_EN	MODE	START / STOP	Reserved	Reserved	KO2	KO1	KO0
0	0	0	0	0	0	0	0

KS_EN This bit master enable bit for the keyboard scanner block. When this bit is "0", the keyboard scanner sequencer will be in the reset state. All other bits in this register will not have function..
0 – disable keyboard sequencer function and put it in power down state
1 – enable keyboard sequencer and accept settings in the Keyboard Scanner Control Register

MODE This bit is used to set the operation mode of the keyboard scanner block and will be valid if KS_EN is set to "1".
0 – Keyboard Scanner Block operates in manual mode (only the KO pin selected by KO[2:0] would be scanned).
1 –Keyboard Scanner Block operates in automatic mode (all 8 KO pins would be activated scanned).

START / STOP This bit is used to start or stop the Keyboard scanning mechanism
0 – Stop keyboard scanning mechanism
1 – This bit is used to start the keyboard scanning mechanism. Setting this bit will start the scanning process. After completion of the scanning, this bit will go back to '0' automatically and key scan will not start until next rising edge of this bit.

KO[2:0] These 3 bits are used to set the KO pin to be scanned in manual operation mode, and the KO data byte to be read by the Read KEY data command (serial communication command).

Bit4, 3 Reserved. Should always write '0' to these bits.

Key Scan Data Register (Register Address : 10H – 17H, RO)

These 8 registers will hold the key scan data

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
KS7	KS6	KS5	KS4	KS3	KS2	KS1	KS0
0	0	0	0	0	0	0	0

8 Serial Interface Format

The serial communication interface in the LCD AND KEYBOARD CONTROLLER is controlled by the SCK and the SD signal. The SCK is an input signal generated from a host device and the SD is a bi-directional open drain pad for data transfer between the host and the LCD AND KEYBOARD CONTROLLER. LCD AND KEYBOARD CONTROLLER is a slave device and the control of the communication is from the external host. 8-bit serial synchronous data format is used during transfer. Each byte of data requires 8-bit data bit and an acknowledge bit. So a total number of 9 clocks signal are required for a byte of data transfer.

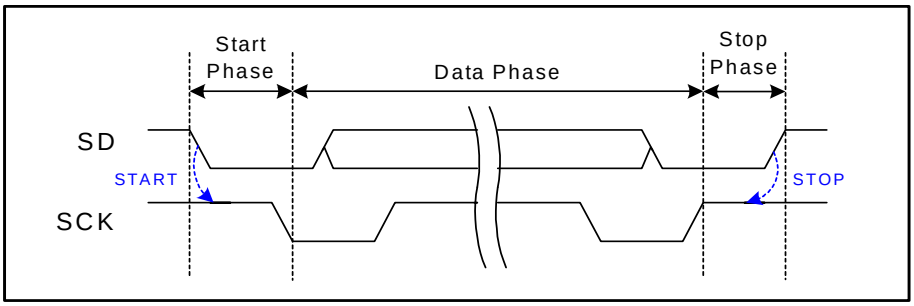


Figure 11: Serial Communication Format

Figure 11 above shows the format of the serial communication data stream. The whole communication can be divided into 3 phases, namely the start phase, the data phase and the stop phase. All the 3 phases of communication are initiated from the host.

When there is no communication between the host and LCD AND KEYBOARD CONTROLLER, both the SD pin and SCK pin will be in the idle state, which is in a logic high level. Communication start by the host generating a start phase signal which is by pulling the SD pin to a logic “0” state while the SCK pin is in a logic “1” level. After the start phase, the host can initiate the data phase transfer. Figure 12 and Figure 13 below shows the protocol for host sending data to LCD AND KEYBOARD CONTROLLER and host requesting data from LCD AND KEYBOARD CONTROLLER respectively. No-matter which situation is, the first byte of data in the data phase is a command byte and must be sent from the host to the LCD AND KEYBOARD CONTROLLER. The command byte will indicate whether data will be sent from the host to the LCD AND KEYBOARD CONTROLLER or the host is requesting data from the LCD AND KEYBOARD CONTROLLER.

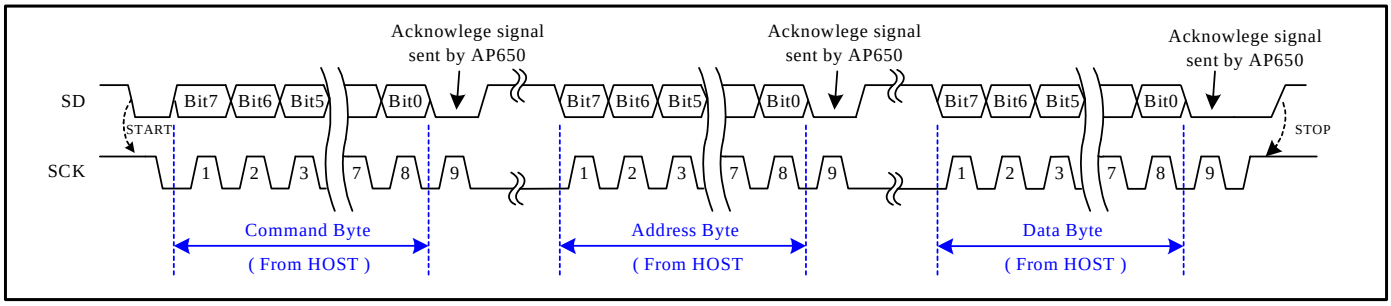


Figure 12: Host transfer data to LCD AND KEYBOARD CONTROLLER

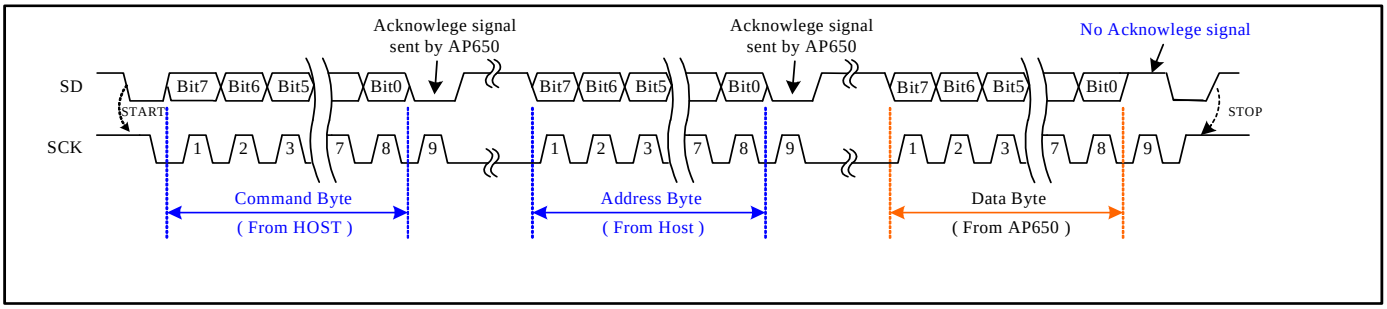


Figure 13: Host Request Data from LCD AND KEYBOARD CONTROLLER

For a host data transfer to LCD AND KEYBOARD CONTROLLER, after each byte of transferred data, the host has to wait for an acknowledge signal sending back by LCD AND KEYBOARD CONTROLLER to signify the successful reception of the data. The acknowledge signal is generated by pulling the SD line to a logic '0" state in the 9th SCK clock. If the host has not received the acknowledge signal, the SD line is in a logic "1" state in the 9th SCK clock, this means that LCD AND KEYBOARD CONTROLLER has not received the data. The host should resend the data until a correct acknowledge signal has received or stopped the whole communication by generating a STOP phase.

A host request data protocol consists of a command byte sending from the host, and then a data phase of data sending back from LCD AND KEYBOARD CONTROLLER to the host. The host has to release the SD line after successfully sending the command byte (an acknowledge signal has received) and then initiate the SCK to receive the data from the LCD AND KEYBOARD CONTROLLER. After 8-bits data have been received by the host, an acknowledge signal should be generated by the host to signify the successful receive of the 8-bit data. Then the SCK can restart again to receive another byte of data. To terminate the stream of bytes, the bus master must no acknowledge the last byte and must generate a STOP condition.

A stop phase is generated by pulling the SD pin from logic "0" to logic "1" state while the SCK pin is in a logic "1" state. The stop phase can be generated at any time to terminate a communication and it will also be used to reset the serial communication interface. So just after power up or the communication link is under some undetermined state, stop phase can be generated to re-initialize the link.

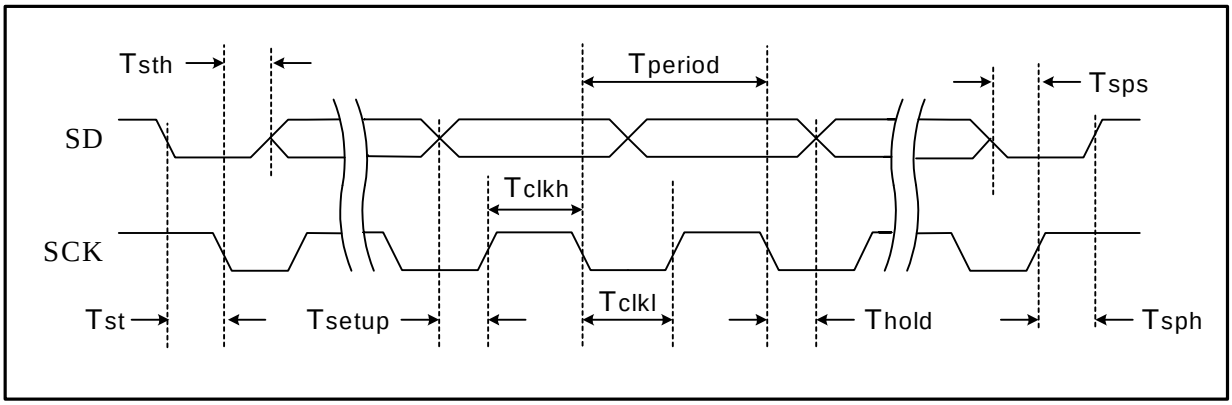


Figure 14: Serial Communication Timing

Symbol	Description	Min	Max	Units
T_{sts}	Start pulse data setup time	100		nS
T_{sth}	Start pulse data hold time	100		nS
T_{sps}	Stop pulse data setup time	100		nS
T_{sph}	Stop pulse data hold time	100		nS
T_{setup}	SD data setup time before SCK clock rising	100		nS
T_{clkh}	SCK high time	500		nS
T_{clkl}	SCK low time	500		nS
T_{period}	SCK clock period	1000		nS
T_{wait}	Wait time after sending a request data command to start receiving data		1	µS

9 Serial Command Sets

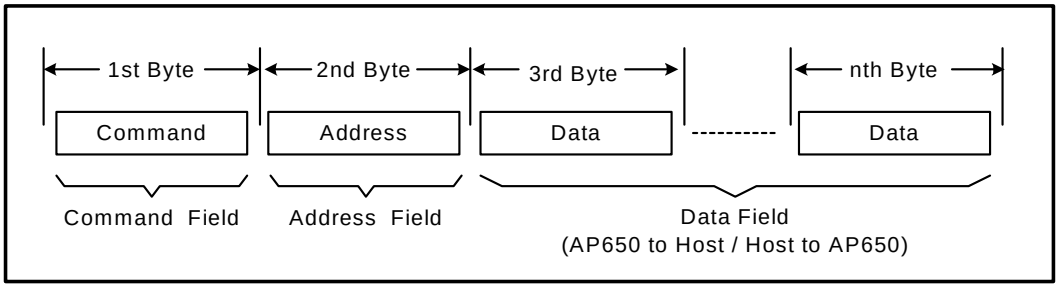


Figure 15: Serial Communication Data Format

Figure 15 above shows the serial communication format between the host and LCD AND KEYBOARD CONTROLLER. The communication is basically divided into 3 portions, namely the command field, the address field and the data field. The command field and the address field are from host and to LCD AND KEYBOARD CONTROLLER while for the data field, it can be from host to LCD AND KEYBOARD CONTROLLER or vice versa. There is no limit for the number of data to be transferred, a stop condition will be used to terminate a communication.

Write Command **ST < 40H > < address > < data1 > < data2 > …. < dataN > SP**

This command is used to write a data to the registers or LCD RAM.

ST Start Sequence

40H This is the command for writing data to LCD AND KEYBOARD CONTROLLER

address RAM This is the address field and can be the address of the internal register or the address of the LCD

data1 … dataN This is the data field.

For a register write operation, the data to be written to the register will be

<data1>	→	<address>
<data2>	→	<address+1>
" "	→	" "
" "	→	" "
<dataN-1>	→	<address+N-2>
<dataN>	→	<address+N-1>

For a LCD RAM write operation, the data to be written to the register will be

<data1, upper nibble>	→	<address>
<data1, lower nibble>	→	<address+1>
<data2, upper nibble>	→	<address+2>
<data2, lower nibble>	→	<address+3>
" "	→	" "
" "	→	" "
<dataN-2, upper nibble>	→	<address+(N-3)*2>
<dataN-2, lower nibble>	→	<address+(N-3)*2+1>
<dataN-1, upper nibble>	→	<address+(N-2)*2>
<dataN-1, lower nibble>	→	<address+(N-2)*2+1>

Read Command **ST < 41H > < address > < data1 > < data2 > …. < dataN > SP**

This command is used to read data from the registers or LCD RAM.

ST Start Sequence

41H This is the command for reading data from LCD AND KEYBOARD CONTROLLER

address RAM This is the address field and can be the address of the internal register or the address of the LCD

data1 … dataN This is the data field.

For a register read operation, the data to be read from the register will be

<address>	→	<data1>
<address+1>	→	<data2>
" "	→	" "
" "	→	" "
<address+N-2>	→	<dataN-1>
<address+N-1>	→	<dataN>

For a LCD RAM read operation, the data to be read from the LCD RAM will be

<address>	→	<data1, upper nibble>
<address+1>	→	<data1, lower nibble>
<address+2>	→	<data2, upper nibble>
<address+3>	→	<data2, lower nibble>
" "	→	" "
" "	→	" "
<address+(N-3)*2>	→	<dataN-2, upper nibble>
<address+(N-3)*2+1>	→	<dataN-2, lower nibble>
<address+(N-2)*2>	→	<dataN-1, upper nibble>
<address+(N-2)*2+1>	→	<dataN-1, lower nibble>

10 LCD Timing Waveforms

The LCD frame frequency FR is ROSCI/512 (64Hz).

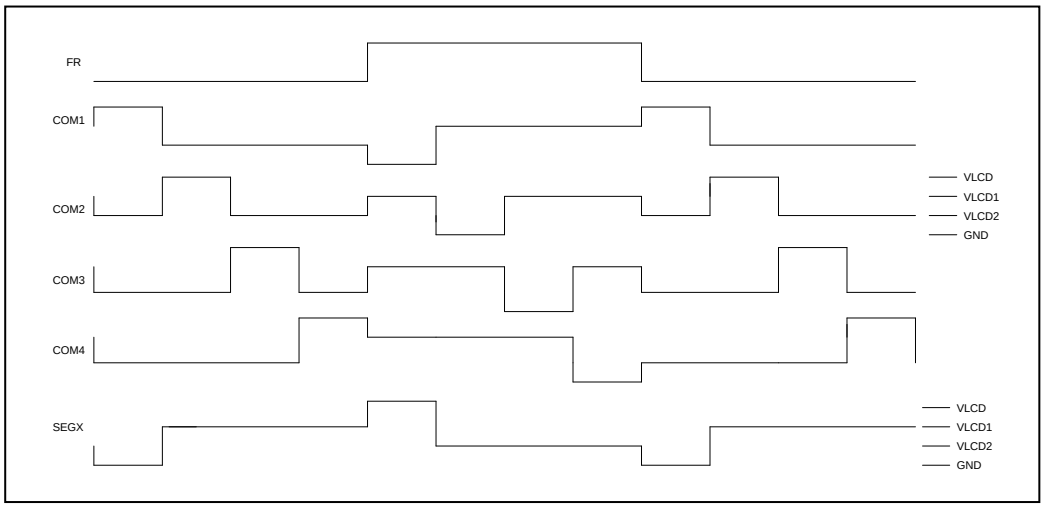


Figure 16: LCD waveforms

11 Electrical Specification

11.1 Absolute Maximum Rating

Under no circumstances the absolute maximum ratings given below should be violated. Stresses exceeding one or more of the limiting values may cause permanent damage to the device.

Parameter	Symbol	Rating	Unit
Power Supply Voltage	VDD	-0.5 to 6.0	V
Input Voltage	Vin	-0.5 to VDD + 0.5	V
Power Dissipation (Ta = 70℃)	Pd	100	mW
Storage Temperature	Tstg	-20 to 125	℃
Operating Temperature	Topr	0 to 70	℃

11.2 Recommended Operating Condition

Parameter	Symbol	Min.	Typ.	Max.	Unit
Power Supply Voltage	VDD	2.0	3.3	5.5	V
Input Voltage	Vin	0	-	VDD	V
Operating Temperature	Topr	0	-	70	℃

11.3 Electrical Characteristics

(VDD=3.0V, Operating temperature = 25℃)

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
VREG	Regulated Supply Voltage	VDD > 2V		1.5		V
VLCD	LCD Supply Voltage	When internal regulator on		VREGx 2		V
		When internal regulator off	VDD		5.5	V
V_{IH}	Input high voltage		$0.8V_{DD}$	-	-	V
V_{IL}	Input low voltage		-	-	$0.2V_{DD}$	V
V_{AIN}	Input voltage for ADC		0		VDD	V
R_{PU}	Pull-up resistance for RESETB	V_{IN} = 0V	-	250k	-	kΩ
R_{PU2}	Pull-up resistance for P3	V_{IN} = 0V	-	100k	-	kΩ
IOL	Output low current for P1 and P2 pins	VOL = 0.4V	-	2	-	mA
IOH	Output high current for P1 and P2 pins	VOH = VDD - 0.4V	-	1	-	mA
Idd	Operating Current	*[1]*		35u		A
IddPd	Power Down current				1	μA

Note:

[1] Assumption :

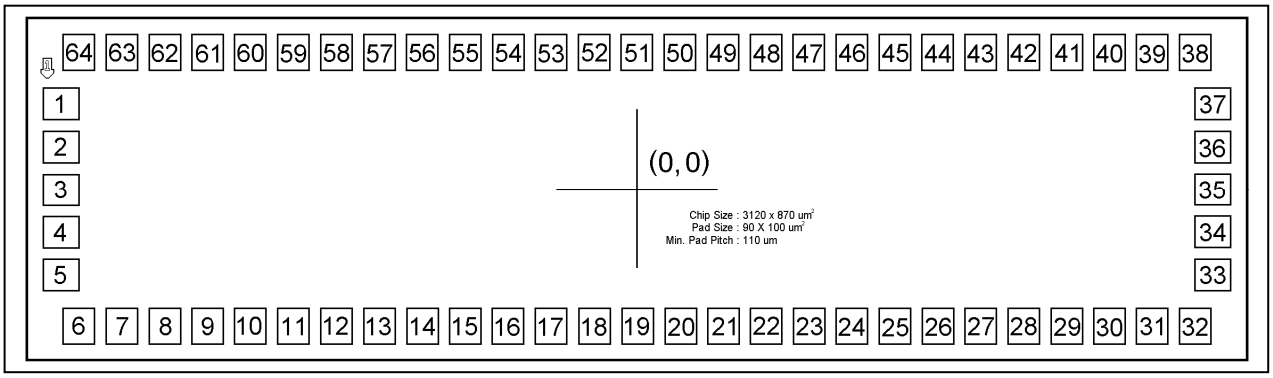
A/D is inactive
No key pressed
Internal LCD supplies are used
Fosc = 32kHz with external R= 950kohm

12 Package Information

12.1 Bonding Information

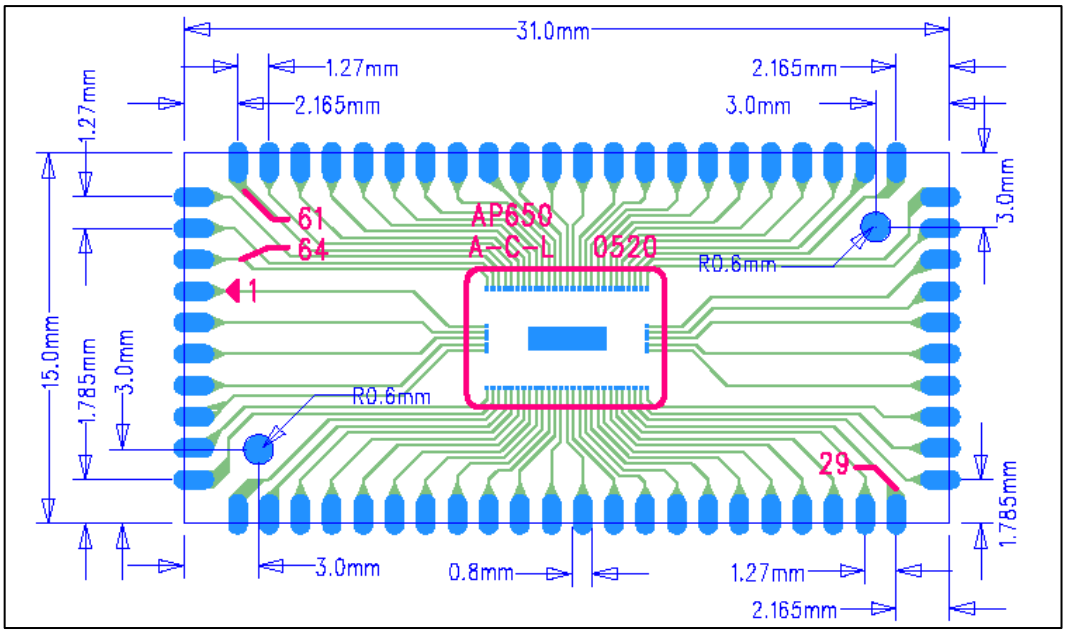
AP650 bonding pad centre coordinates and pin assignment.

Die Size : 3120 x 870μm²
Pad Size : 90x100μm²
Min.Pad Pitch : 110μm

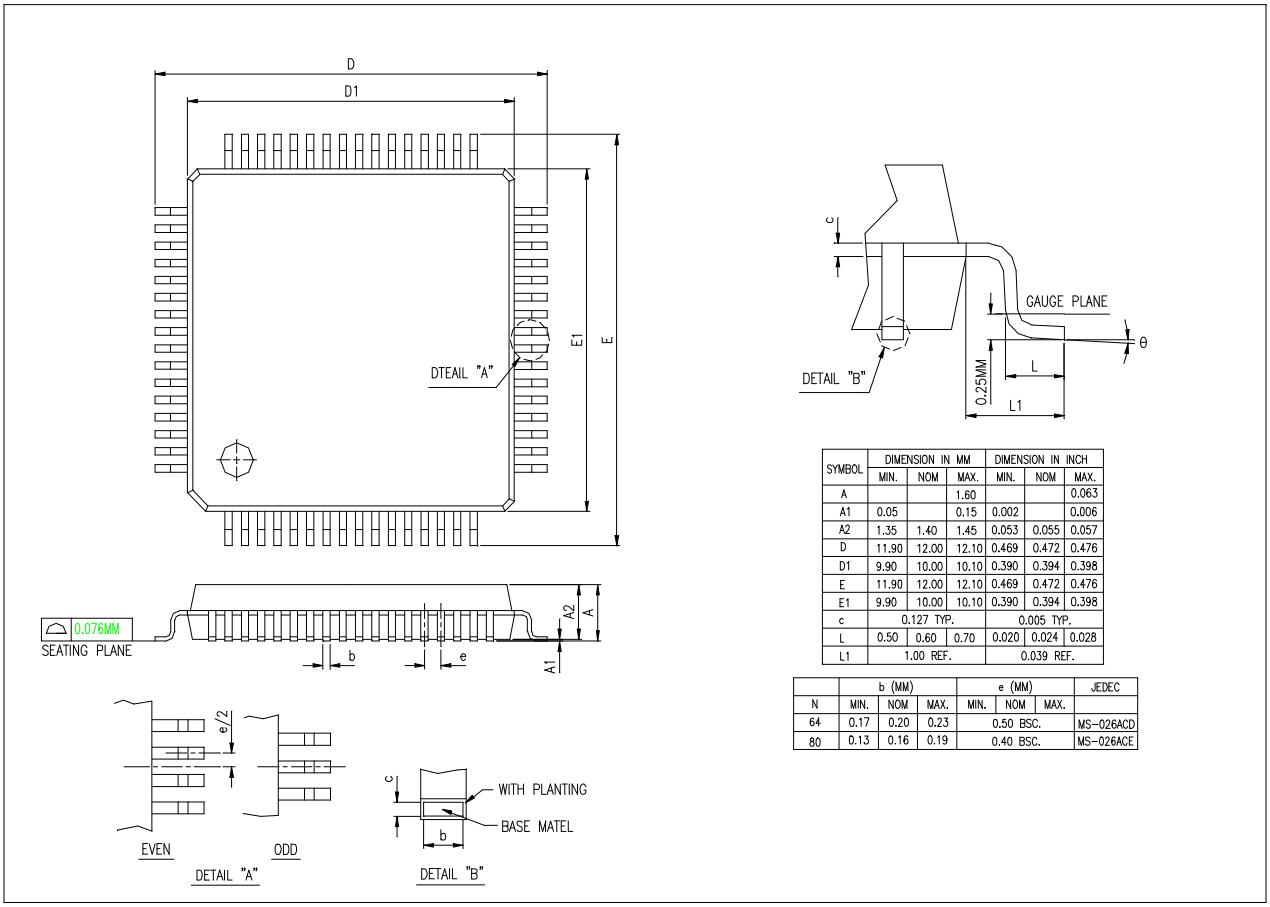


PAD #	PAD NAME	X- coordinate (μm)	Y-coordinate (μm)		PAD #	PAD NAME	X- coordinate (μm)	Y-coordinate (μm)
1	SEG18	-1476	220		33	SCLK	1476	-220
2	SEG19	-1476	110		34	SDATA	1476	-110
3	SEG20	-1476	0		35	RESETB	1476	0
4	SEG21	-1476	-110		36	VDD	1476	110
5	SEG22	-1476	-220		37	VREG_OUT	1476	220
6	SEG23	-1430	-351		38	C1A	1430	351
7	SEG24	-1320	-351		39	C2A	1320	351
8	SEG25 / P10	-1210	-351		40	VLCD	1210	351
9	SEG26 / P11	-1100	-351		41	VLCD1	1100	351
10	SEG27 / P12	-990	-351		42	VLCD2	990	351
11	SEG28 / P13	-880	-351		43	GND	880	351
12	SEG29 / P14	-770	-351		44	COM1	770	351
13	SEG30 / P15	-660	-351		45	COM2	660	351
14	SEG31 / P16	-550	-351		46	COM3	550	351
15	SEG32 / P17	-440	-351		47	COM4	440	351
16	SEG33 / P20	-330	-351		48	SEG1	330	351
17	SEG34 / P21	-220	-351		49	SEG2	220	351
18	SEG35 / P22	-110	-351		50	SEG3	110	351
19	SEG36 / P23	0	-351		51	SEG4	0	351
20	SEG37 / P24	110	-351		52	SEG5	-110	351
21	SEG38 / P25	220	-351		53	SEG6	-220	351
22	SEG39 / P26	330	-351		54	SEG7	-330	351
23	SEG40 / P27	440	-351		55	SEG8	-440	351
24	ADC0 / P30	550	-351		56	SEG9	-550	351
25	ADC1 / P31	660	-351		57	SEG10	-660	351
26	ADC2 / P32	770	-351		58	SEG11	-770	351
27	ADC3 / P33	880	-351		59	SEG12	-880	351
28	ADC4 / P34	990	-351		60	SEG13	-990	351
29	ADC5 / P35	1100	-351		61	SEG14	-1100	351
30	ADC6 / P36	1210	-351		62	SEG15	-1210	351
31	ADC7 / P37	1320	-351		63	SEG16	-1320	351
32	ROSC1	1430	-351		64	SEG17	-1430	351

12.2 COB Package



12.3 LQFP-64



SYMBOL	DIMENSION IN MM			DIMENSION IN INCH		
	MIN.	NOM	MAX.	MIN.	NOM	MAX.
A			1.60			0.063
A1	0.05		0.15	0.002		0.006
A2	1.35	1.40	1.45	0.053	0.055	0.057
D	11.90	12.00	12.10	0.469	0.472	0.476
D1	9.90	10.00	10.10	0.390	0.394	0.398
E	11.90	12.00	12.10	0.469	0.472	0.476
E1	9.90	10.00	10.10	0.390	0.394	0.398
c	0.127 TYP.			0.005 TYP.		
L	0.50	0.60	0.70	0.020	0.024	0.028
L1	1.00 REF.			0.039 REF.		

	b (MM)			e (MM)			JEDEC
N	MIN.	NOM	MAX.	MIN.	NOM	MAX.	
64	0.17	0.20	0.23	0.50 BSC.			MS-026ACD
80	0.13	0.16	0.19	0.40 BSC.			MS-026ACE

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